DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 19: Random Walks

The Simple Symmetric Walk · The $\sqrt{n}$ Law · Martingale Properties

September 16, 2026

Today's Agenda

1. The simplest random process

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4. Martingale properties
5. Connection back to the binomial tree

Learning Objectives

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- State and explain the $\sqrt{n}$ scaling law.
- Show S_n and $S_n^2 - n$ are martingales.
- Relate the walk to Week 6's binomial tree.

The Wandering Pedestrian

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Stand at a lamppost; every second flip a fair coin and pace east or west. After a minute, on average you've gone *nowhere*. But you're almost certainly not at the lamppost either. The interesting question: *how far* have you wandered? Answer: like $\sqrt{n}$, not n.

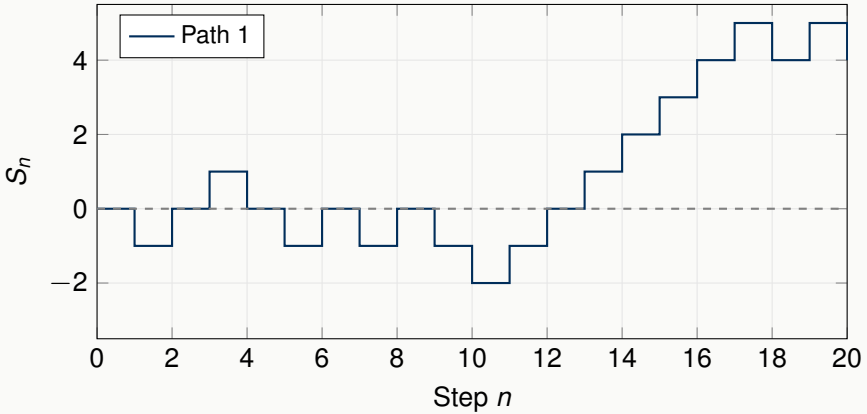
Definition

Simple Symmetric Random Walk

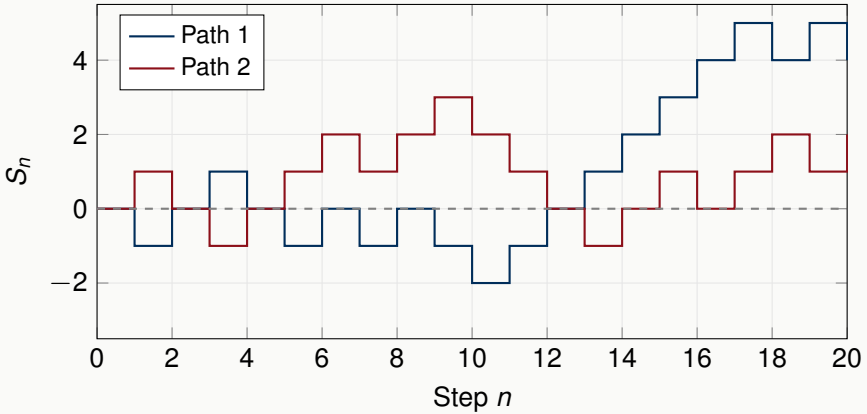
$X_1, X_2, \ldots$ i.i.d. with $\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$.

$$S_0 = 0, \qquad S_n = \sum_{i=1}^{n} X_i$$

Two Sample Paths



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Mean and Variance

Key Takeaway

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$\mathbb{E}[X_i] = 0$, so $\mathbb{E}[S_n] = \sum \mathbb{E}[X_i] = 0$. $\mathbb{E}[X_i^2] = 1 \Rightarrow \mathrm{Var}(X_i) = 1$; **independence** $\Rightarrow$ variances add:

$$\mathrm{Var}(S_n) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = n$$

Independence Is Doing Real Work

Remark

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Variances add only because steps are **independent** – all $\mathrm{Cov}(X_i, X_j)$ vanish. If successive steps were correlated (momentum), $\mathrm{Var}(S_n)$ would grow *faster* than n. Everything built on $\mathrm{Var}(S_n) = n$ rests on this.

The $\sqrt{n}$ Law

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Typical distance grows like $\sqrt{n}$ – **not** n.

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200,000 simulated walks

	n=1	n=4	n=25	n=100
Var (sim)	1.00	3.98	24.93	100.03
Var (theory)	1	4	25	100
sd (theory)	1	2	5	10

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The Geometric Middle Ground

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You've used "annualize by $\sqrt{252}$" since Week 3 – here's the derivation.

Two Martingales From One Walk

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Since $S_{n+1} = S_n + X_{n+1}$ with $\mathbb{E}[X_{n+1}] = 0$:

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + 0 = S_n \checkmark$$

The Second Martingale

Expand $S_{n+1}^2 = S_n^2 + 2S_nX_{n+1} + X_{n+1}^2$, use $\mathbb{E}[X_{n+1}] = 0$, $\mathbb{E}[X_{n+1}^2] = 1$:

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Squared displacement grows at exactly 1 per step.

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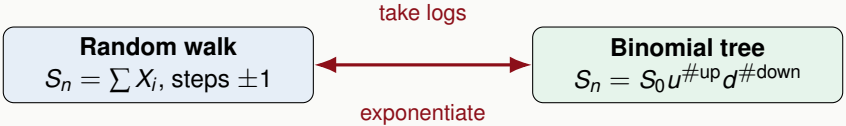
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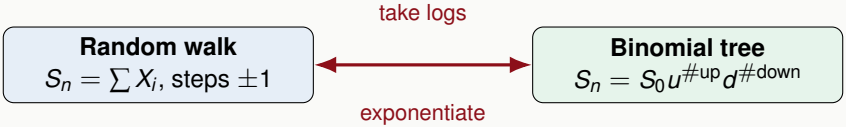
Why this matters

Squared displacement grows at exactly 1 per step. This is the ancestor of $W_t^2 - t$ (Lecture 21) and ultimately of $(\mathrm{d}W)^2 = \mathrm{d}t$ in Week 8.

The Bridge: Take Logarithms



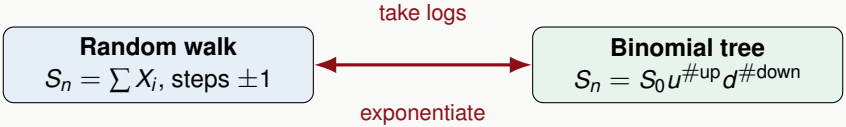
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Why did CRR set $d = 1/u$? In log space that's $\ln d = -\ln u$ – down-steps are exactly the negatives of up-steps. Precisely the *symmetric* random walk. What looked like convenience in Lecture 16 was a statement about the underlying walk.

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- $\sqrt{n}$ **law**: displacement grows like $\sqrt{n}$ – origin of $\sqrt{\text{time}}$ volatility scaling.
- S_n and $S_n^2 - n$ are **martingales**.
- A CRR binomial price is an **exponentiated random walk**.

Check Your Understanding

Discuss (2 minutes)

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2. Annual $\sigma = 24\%$: estimate monthly and daily volatility.
3. Asymmetric walk with $\mathbb{P}(X_i = +1) = 0.6$: still a martingale?

Looking Ahead

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Scaled Limits – Lecture 20

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Scaled Limits – Lecture 20

The bridge from discrete to continuous time.